

AI Chatbots in K-12 Education: An Experimental Study of Socratic vs. Non-Socratic Approaches and the Role of Step-by-Step Reasoning

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Highlights

- An experiment was conducted where K-12 students (N=122) used AI chatbots to solve school-related problems.
- Two interventions were tested: (1) comparing AI-generated step-by-step explanations to predictions alone, and (2) contrasting AI chatbot support inspired by the Socratic method (guided questioning) versus non-Socratic (direct answers) support.
- Participants performed better under step-by-step explanations. They also engaged more frequently with the Socratic AI, though this approach did not result in improved performance and retain of knowledge compared to the non-Socratic AI.
- While students found interactions with the AI useful, they perceived the Socratic AI as less helpful than non-Socratic overall.
- The findings highlight challenges in designing AI tutors that effectively foster critical thinking while maintaining student satisfaction, raising concerns about their adoption in educational settings.

Abstract

How does integrating large language models (LLMs) into classroom activities influence students' learning? To investigate this question, we conducted a randomized experiment with students aged 14-16 (N=122), testing two interventions: (i) comparing AI-generated solutions with step-by-step explanations versus solutions alone and (ii) Socratic AI chatbots promoting critical thinking through incremental guidance versus non-Socratic AI offering direct solutions. Students completed school-related tasks under these conditions. We collected students' performance and interaction logs data, as well as their attitudes through self-reported surveys. Results indicated that AI-generated explanations significantly improved students' performance over solutions alone, highlighting the benefits of step-by-step guidance. The Socratic AI approach fostered significantly greater engagement and interaction. However, it did not achieve significant improvements in learning, and a higher fraction of students perceived it as less helpful. Furthermore, despite overall positive perceptions of AI assistance, students exhibited limited retention, failing to apply learned concepts to new situations without the aid of AI. These findings underscore both the potential and limitations of LLM-based chatbots in K-12 education, especially in balancing the development of critical thinking with user satisfaction.

Keywords: artificial intelligence; large language models; learning; education policy; experiment; k-12

1 Introduction

Integrating artificial intelligence (AI) chatbots into educational settings has been debated for over three decades (Roll and Wylie 2016). Recent advances in generative AI and the rise of large language models (LLMs) have accelerated this integration, opening new possibilities for improvements across several academic tasks (Yan, Greiff, et al. 2024; Kasneci et al. 2023; Tlili et al. 2023; Debets et al. 2025), including problem-solving (Urban et al. 2024), personalised assistance (Darvishi et al. 2024), language learning (Derakhshan and Ghiasvand 2024; Song and Song 2023), and evaluating students' work (Henkel et al. 2024). However, the impact of these applications on students' learning remains under scrutiny (Farrokhnia et al. 2024; Weidinger et al. 2022; Walczak and Cellary 2023). Most current AI chatbots emphasize convenience and efficiency, offering quick answers or summaries, rather than engaging students in deeper learning processes such as reasoning, reflection, or explanation. As a result, there is growing interest in exploring how specific forms of AI-generated instructional support can enhance student learning and critical thinking.

In this study, we investigate two such forms: (1) AI-generated step-by-step explanations, and (2) Socratic dialogue delivered by an AI tutor. Although distinct in interaction style, both interventions share a common pedagogical goal: to enhance students' thinking by making reasoning explicit and guiding them through problems rather than simply delivering solutions.

Integrating step-by-step reasoning into LLMs represents a major advancement over earlier AI systems. Simple prompts like "let's think step by step" have been shown to significantly improve LLMs performance across a range of accuracy benchmarks (Kojima et al. 2022). Beyond accuracy, adding explanations enhances the interpretability of AI outputs, which is essential for enhancing human trust in AI outputs (Rudin et al. 2022; Rai 2020; Ferrario and Loi 2022). At the same time, the same models are also prone to generating inaccurate solutions and flawed reasoning (Saxton et al. 2019; Kalyan et al. 2021; Hendrycks et al. 2021), which can undermine trust and, ultimately, inhibit learning.

In educational settings, explanations have been found to promote children's learning and support development of scientific reasoning, particularly among K-12 students (DeJong and Mooney 1986; Legare 2014; Danovitch et al. 2021). However, they are not always effective, as children sometimes bypass instructions and focus solely on the solutions without engaging with the reasoning provided (Wittwer and Renkl 2008). Since most AI chatbots are designed to offering solutions alongside their explanations, it remains an open question how these configurations affect students' engagement with the reasoning process and, ultimately, their learning outcomes.

This leads to our first two research questions (RQs):

- **RQ1:** Do AI-generated explanations enhance students' problem-solving performance in school tasks?
- **RQ2:** How do AI-generated explanations influence students' perceived credibility of AI-generated solutions?

Socratic dialogue offers a complementary approach by encouraging students to generate their own reasoning through guided questions, thereby activating critical thinking and metacognition (Lam 2011; Dalim, Ishak, and Hamzah 2022; Wilberding 2021; Cleveland 2015). This approach encourages *active* learning, which can result in better understanding and retention (Vygotsky and Cole 1978; Corno and Mandinach 1983; Chi and Wylie 2014). Moreover, encouraging students to find their own answers, rather than relying on provided solutions, boosts their confidence in reasoning (Cleveland 2015). Yet, Socratic methods are in contrast to most existing AI chatbots, such as ChatGPT, that focus on providing direct solutions, without necessarily prompting students to deep reflection. This leads to the following research questions:

- **RQ3:** Do student-AI interactions guided by the Socratic method promote better performance?
- **RQ4:** Do student-AI interactions guided by the Socratic method enhance students' confidence in their answers?

Beyond learning outcomes, exploring the trade-offs between Socratic and non-Socratic AI raises an important question about student *satisfaction*. If students find interacting with Socratic AI chatbots unengaging or frustrating, such perceptions might drive students away from Socratic AI to seek other (non-Socratic) tools. Indeed, research on conversation length with LLM-powered chatbots has shown that people tend to have mixed reactions when chatbots are instructed to engage in more extended conversations (Huang et al. 2024). Therefore, aligning educational benefits with user satisfaction is crucial for ensuring the long-term adoption of AI learning tools, leading to our last research question:

- **RQ5:** How do students perceive the helpfulness of Socratic AI compared to non-Socratic AI?

To investigate these questions, we conducted a randomised experiment with k-12 students ($n = 122$). Participants completed academic tasks with the assistance of an AI tutor powered by GPT-4. The experiment employed two independent interventions: (i) Explanation vs. No Explanation – comparing students' performance and perceptions when shown a solution with versus without AI-generated step-by-step reasoning; and (ii) Socratic vs. Non-Socratic Dialogue – comparing AI tutors that provided direct answers versus those that engaged students through open-ended questioning.

The first intervention allows us to isolate the effects of transparent reasoning on student accuracy and trust in AI-generated solutions. This is particularly important in K–12 contexts, where students may not possess the background knowledge to critically evaluate AI output, and where interpretability may influence both understanding and overreliance. The second intervention examines the trade-off between engagement and efficacy in dialogue-based AI tutoring. While the Socratic method has a long history of fostering deep learning, its AI implementation in current AI tools raises new questions about usability, perceived helpfulness, and the role of trust in human-AI interaction, which our intervention allows us to investigate.

Our study contributes to a growing debate about pedagogical alignment and how pedagogical principles and techniques, like the Socratic approach, should be used to integrate AI in education. For instance, Khan Academy has developed its AI tutor, Khanmigo, based on Socratic principles, aiming to foster criti-

cal thinking and engagement in learners.¹ Several commentators have argued that the Socratic method is particularly effective for designing AI tutors for children, as it enhances critical thinking and discourages students from copying AI-generated responses without scrutiny (Lara and Deckers 2020).² Beyond education, Socratic AI systems are also being developed to foster critical thinking among the general public, particularly in efforts to combat disinformation (Duelen, Jennes, and Van den Broeck 2024).

¹<https://docsbot.ai/prompts/education/khanmigo-lite-socratic-tutor>

²See also <https://research.gatech.edu/ai-oral-assessment-tool-uses-socratic-method-test-students-knowledge>

2 Relevant Background

Our analysis of integration of AI chatbots into educational contexts draws on intersecting theoretical and empirical foundations from learning sciences, human-computer interaction, and AI interpretability.

2.1 Theoretical Considerations

Explanations are a central component of learning, both in everyday cognition and formal education. Research suggests that generating explanations, especially self-explaining, can be more effective for learning and generalization than simply reading text or receiving feedback (Lombrozo 2006). In educational settings, explanations are widely used across contexts, such as classroom instruction (Ma 1999), tutoring (Chi et al. 2001), peer learning (Webb 1989), and learning from texts (Goldman 2004), but their effectiveness depends heavily on how they align with learners’ prior knowledge, focus on conceptual understanding over rote procedures, support active learning, and do not replace the learner’s own knowledge construction (Wittwer and Renkl 2008).

AI-generated explanations, however, often fall short on these dimensions: they may not adapt to individual knowledge levels, ignore learner agency, and risk fostering overreliance due to the opaque, “black-box” nature of models, potentially spreading misinformation or suppressing critical thinking (Rudin et al. 2022). Nonetheless, chatbots offer a unique opportunity for adaptivity, allowing learners to engage in Socratic dialogue and refine their understanding through follow-up questions (Yan, Greiff, et al. 2024). Our study explores this duality by first examining the impact of non-adaptive AI explanations and then testing whether adaptive, Socratic-style interactions better support learning compared to traditional classroom explanations.

Building on this, adaptive learning via AI chatbots can be conceptualized as a form of *collaborative learning* between the student and the AI tutor (Dillenbourg 1999). Collaboration, as defined by Dillenbourg, encompasses four aspects: the learning situation, the nature of interactions, the underlying cognitive mechanisms, and the outcomes for learners. In this study, we focus on the second aspect—the *interactions*—to investigate how different dialogic strategies employed by a LLM shape students’ learning experiences and perceptions. By framing the AI-student exchange as a collaborative process, we aim to understand how the structure and responsiveness of chatbot explanations, especially those modeled on Socratic questioning, can promote deeper engagement, conceptual understanding, and critical reflection compared to more static or directive approaches.

2.2 The role of LLMs in Education

A growing body of research highlights the tremendous potential of LLMs to improve student learning outcomes. A recent scoping review identifies 53 distinct use cases for LLMs in education (Yan, Sha, et al. 2024), ranging from automatic grading and personalised feedback to teaching support and content generation, reflecting the high versatility and scalability of LLMs. Within this expanding literature, this study centres on *AI chatbots*, conversational agents powered by LLMs, and their educational applications,

reflecting their increasing prominence and potential impact in this field.

Early investigations into the use of AI chatbots, such as ChatGPT, in educational settings have identified both opportunities and challenges (Farrokhnia et al. 2024; Noroozi et al. 2024). On the one hand, ChatGPT can serve as a personalised tutor, assist in drafting assignments, or generate creative prompts for classroom activities. On the other hand, concerns have been raised around academic integrity, overreliance on AI-generated content, and the need for critical thinking skills when interpreting chatbot outputs.

A recent review of over 70 studies provides a comprehensive overview of how chatbots are used in various educational contexts (Debets et al. 2025). It classifies chatbot applications by roles, including tutoring, giving feedback, and helping with administrative tasks, and examines how they are designed and utilised. The review finds that AI chatbots generally meet their goals. For example, a quasi-experimental study with 320 middle schoolers found that an AI teaching chatbot significantly improved math knowledge (Xing et al. 2025). Similarly, a meta-analysis of 24 RCTs shows that AI chatbots significantly enhance student outcomes, with stronger effects in higher education (R. Wu and Yu 2024). However, most studies are short-term, may be subject to publication bias, and often lack a strong theoretical foundation.

Despite promising results, key limitations remain. For instance, Er et al. (2024) found that students in a programming class performed better with instructor feedback, which was seen as more useful and fair than AI-generated feedback, highlighting the current gap between AI and human instruction. Similarly, evidence of performance boosts from non-education settings is mixed. A meta-analysis by Vaccaro, Almaatouq, and Malone (2024) found that human-AI teams often performed worse than either humans or AI alone, suggesting a more nuanced relationship. In education, additional concerns are that integrating AI might encourage cheating, over-reliance, or superficial understanding (Farrokhnia et al. 2024; Yusuf, Pervin, and Román-González 2024; Eke 2023; Lee et al. 2024), a risk that is difficult to evaluate without longitudinal studies.

The effectiveness of AI in education often depends on students' attitudes and perceived usefulness of AI tools, understanding which can guide curriculum design. For instance, Li et al. (2024) found that K–12 students' views on AI tools were linked to factors like readiness, confidence, and anxiety. Similar concerns have been noted from teachers' perspectives as well (Kaplan-Rakowski et al. 2023). Our study builds on this by exploring how interaction styles, such as Socratic vs. non-Socratic, shape students' perceptions of AI helpfulness.

This study advances the literature on human-AI collaboration in educational contexts by examining the role of step-by-step reasoning and the impact of different modes of AI interaction. Additionally, it contributes to ongoing research exploring the factors that influence students' engagement with AI. Gaining insights into the value these combinations bring is crucial for developing practical guidelines for schools and educational institutions.

Most existing research focuses on general AI chatbots, like ChatGPT, which are not specifically designed for students and often lack clear pedagogical foundations. Our study highlights the importance of purposeful design and pedagogical goals, contributing to the growing field of AI chatbot design for education or design for learning (Gašević, Siemens, and Sadiq 2023). This research calls for a multidisciplinary ap-

proach combining pedagogy and human-computer interaction. Building on recent efforts ([Dai et al. 2023](#); [Yan, Sha, et al. 2024](#); [Joksimovic et al. 2023](#)), we examine how k–12 students interact with different AI chatbot styles. By doing so, the study underscores both the potential and challenges of applying classical pedagogical methods, such as the Socratic approach, emphasizing the need to provide explanations rather than just answers.

3 Methods

We conducted a field experiment across two schools to investigate the impact of different AI tutoring strategies on students' critical thinking, self-confidence, and task performance. Participants engaged with a custom-designed web application, the *AI Tutor*, which randomly assigned students to interact with distinct versions of GPT-4-based chatbots during structured educational activities. The experimental design featured two primary manipulations: (1) the presence of step-by-step reasoning by the AI and (2) the use of Socratic versus non-Socratic dialogue styles. To assess the effects, we employed objective performance measures alongside pre- and post-task surveys capturing students' cognitive and affective responses. The following subsections detail the AI Tutor system, participant characteristics, and experimental procedures.

3.1 AI Tutor

To assess the impact of LLMs on students' critical thinking, we developed an *AI Tutor*, a web-based application built using Python with the Flask framework. The application integrates OpenAI APIs, allowing students to interact with the GPT-4.0 model while performing different educational tasks. This app's key features include:

- **User authentication:** A secure and anonymous login system for students.
- **Web Survey:** A web interface to collect data from students, including answering survey questions and performing simple tasks, such as writing a short essay or solving simple math problems.
- **AI Chatbot:** A chatbot powered by GPT-4 provided students with personalised assistance and tutoring during the session. A new chatbot instance was associated with each question for each student, isolating each conversation from interactions in previous questions. The chatbot was available to students only for specific questions or tasks under the researchers' control.
- **Data logging:** A SQL database storing students' conversation logs with the AI chatbot, including texts, timestamps, and survey responses.
- **Randomised assignment:** A system to randomly assign registered students to different versions of the AI chatbot or treatment groups to explore the causal impact of various configurations on students' interactions and performance.

3.2 Participants

We recruited N=122 students between 14 and 16 years old enrolled in secondary education from two schools: one located in Brussels, Belgium, and the other in Seville, Spain.³ The experiment was conducted during school hours at the participating schools (Fig. 1). Both schools are bilingual secondary institutions that follow a European curriculum framework, with English as the primary language of instruction across most subjects. The student populations at both schools are culturally and linguistically diverse,

³We recruited schools in different countries to increase the external validity of our study. The country choice was by convenience as the authors lived in Brussels and Seville at the time of the experiment.

including a mix of local and expatriate families. Many students come from middle- to upper-middle-class socioeconomic backgrounds and are accustomed to using digital technologies in their academic work: both schools are equipped with computer laboratories and digital interactive whiteboards for classroom instruction.

All participants had appropriate levels of English proficiency, given the language policy of the schools. Even so, the AI Tutor was multilingual, allowing students to interact with it either in English or in their native language. The experimental instructions were in English to ensure consistency across sites. However, if needed, students could translate the instructions using the browser's automatic translation service.

3.2.1 Ethical Approval and Data Privacy

We recruited students after receiving ethical approval from the European Commission's internal Ethical Review Board, ensuring that the consent procedures, data protection requirements, and experimental protocol complied with local laws and ethical research standards.⁴ The data privacy protection protocol was approved by the European Commission's data protection officer. As an additional safeguard, given the minor age of the participants, we required their parents or legal guardians to provide us with written consent, allowing their children to participate in the study. The students also had to give their assent to participate by completing an online consent form.

3.3 Experimental sessions

About ten experimental sessions were conducted at the schools between November 11 and 12, 2023, in Brussels and February 8 and 9, 2024, in Seville. Each session took about two hours. In the first 45-60 minutes, students received a personal computer and were asked to perform three main tasks using the AI tutor and answer a questionnaire without the AI tutor. In the following 45-60, there was a group discussion on how the students judged the interactions with the AI tutor and a more general debate on how they perceived the potential benefits and drawbacks of integrating LLMs in the classroom. The results of the group discussions are not discussed in this paper; however, they were used as material for interpreting the results of the experimental study.

3.4 Experimental Conditions

Figure 2 illustrates the two randomised manipulations in our experimental design: (1) AI step-by-step reasoning and (2) Socratic vs non-Socratic AI. It is important to note that these were conducted as independent experiments, with different dependent and independent variables, rather than a factorial 2x2 design.

⁴Clearance obtained on May 17, 2023.

3.4.1 AI step-by-step reasoning

The first manipulation focuses on a task in which students are asked to estimate the value of coins in a jar (see Section A.1 for the details). Specifically, we used the coin jar from Steiner’s experiment (Steiner 2015), aimed originally at assessing Internet users’ guessing accuracy. For this intervention, we varied the AI’s response by providing either a complete answer, which included both an estimated value and a step-by-step explanation generated by the AI tutor, or a partial answer that provided only the estimate without additional details. In both conditions, all participants received the same estimated value (\$213), but those in the full-answer condition also viewed an explanation outlining the step-by-step reasoning the AI used to arrive at this estimation based on the jar image.⁵

This type of task falls under the category of open-ended estimation problems, often referred to as *Fermi problems*, which are widely used in education and research. A growing research highlights their pedagogical value: they help students tackle complex problems in STEM classes (Arleback and Albarracin 2019) and foster students’ motivation (Abay and Filiz 2020). For example, Micklo (1999) describes classroom estimation activities involving marbles, closely resembling our task, as a way to develop students’ numerical reasoning, while pedagogical networks, such as EU Math Excellence for Upper Secondary Education, recommend the use of these tasks to improve students’ mathematical reasoning and problem-solving skills.⁶ In research, Ferrando and Albarracin (2021) explores problem-solving skills of 8-16 years old students estimating how many people that can fit in their school playground; a task conceptually similar to estimating the value of coins in a jar. In behavioural economics, similar estimation exercises are often used to illustrate the winner’s curse (Thaler 1988). Overall, the task aligns well with established educational practices and research.

This setup allows us to examine how AI-provided explanations affect students’ performance and their perceptions of the AI’s accuracy. Specifically, we focus on three outcome variables: (1) the propensity to update their initial guess, (2) the accuracy of students’ final estimations, measured as the (absolute) difference between their final guesses and the actual coin value, and (3) students’ perceived accuracy of the AI, rated on a scale from low to high. We also asked participants (4) to rate the perceived accuracy of the mean guess among 600 people (\$596), which exaggerated the correct value, as reported in the original article (Steiner 2015).⁷

3.4.2 Socratic vs Non-Socratic AI

The second manipulation involved randomly assigning students to one of two different types of AI tutors: Socratic or non-Socratic AI tutors. Both tutors were powered by the same underlying large language model (GPT-4). Still, each was instructed with different “system messages” to create different behaviours,

⁵Using GPT 4.0, we uploaded the image of the coin jar asking for an estimate of the value of coins ten times. We then selected the median response for the experiment.

⁶See: <https://excellenceinmath.eu>

⁷Although this task involves estimation (participants are expected to apply logical reasoning and draw on relevant knowledge, rather than rely solely on intuition), we acknowledge that the instruction to ‘guess’ might have unintentionally suggested that reasoning was not required. While we believe this interpretation is unlikely, if some participants did perceive the task this way, it could represent a potential confounding factor in our analysis.

as illustrated in Table 1.⁸ For the Socratic tutor, the system message asked the model to engage students with open-ended, thought-provoking questions, encouraging them to think critically about their responses. In contrast, the non-Socratic AI tutor was instructed to provide concise, direct answers without necessarily engaging in deeper dialogue or posing further questions.

This setup allows us to investigate the impact of different pedagogical AI tutoring approaches on students' performance and perceptions. Specifically, we asked students to use the AI tutor while performing three tasks: guess an unknown quantity ("How much water in litres do students consume at our school each week?"), express an opinion and write a short essay ("What is your opinion about the effect of social media on teenagers?"); respond to physics questions on how sound propagates in different media ("In which of the following materials does sound travel faster?"). These questions enabled us to assess the AI tutor's impact on students' learning and problem-solving abilities.

We focused on two primary metrics: (1) confidence in their answers and (2) perceived usefulness of interacting with the AI tutor. Specifically, students were asked, "How confident are you that the answer you provided is accurate?" on a five-point scale ranging from "not confident at all" to "very confident." They were also asked, "How helpful was it to interact with the AI tutor?" This was also rated on a five-point scale, from "Not at all helpful" to "Very helpful." Only for task 3, we had an additional metric, which was the correctness of their answers.

3.5 Student Characteristics

To examine student interactions with AI and their potential impact on learning and critical thinking, we collected a range of self-reported and behavioural data. The full questionnaire is provided in the Appendix (Section E).

3.5.1 AI Attitudes and Usage

To better understand students' general attitudes toward AI, we selected four questions from Schepman and Rodway's AI Attitudes Scale (Schepman and Rodway 2020). These questions explored students' beliefs about AI's societal benefits, potential dangers, capacity to support learning, and the likelihood of misuse by students (Appendix, Section E). We also asked students about their direct experience using ChatGPT for homework — the most common chatbot at the time of the study — and their beliefs about how many of their peers use ChatGPT for their homework, thereby capturing both individual behaviour and the perceived social norm around AI usage. These measures help establish baseline orientations that may moderate how students engage with and evaluate AI explanations (related to RQ-1) and how they perceive different AI interaction styles (related to RQ-3).

⁸An AI's system message guides how the AI interprets the conversations by setting parameters for interaction.

3.5.2 AI Academic Habits and Skills

Academic skills or habits may moderate the effect of AI interactions on problem-solving activities. Accordingly, to account for participants' heterogeneity in academic skills, we asked students to report their average school grades within five categories. We also asked students about their academic habits or the challenges they face at school, such as how often they complete their homework assignments on time and what factors affect their ability to do so.

3.5.3 Student-AI Interaction Metrics

To quantify engagement, we analyzed students' interaction logs with the AI Tutor, recording the number of turns and the word counts as a proxy for interaction intensity. While this provides a basic behavioural metric, it does not capture the quality of cognitive engagement or helpfulness, limiting its direct relevance to RQ-1 and RQ-2. Therefore, we also asked students about how useful they found the AI interactions and how confident they were in their answers to selected tasks. These self-reported metrics are needed to assess whether explanations were helpful to students (for RQ-1) and whether Socratic dialogue promoted critical thinking or confidence (for RQ-2). To complement these measures, logs of student-AI interactions for selected questions are included in the Appendix.

4 Results

4.1 Overview of Student Demographics

A total of 122 students participated in the study, 64 in Brussels and 58 in Seville. The sample was gender balanced. Over half of the students reported B+ grades and prior use of ChatGPT, with boys more likely to report ChatGPT experience. Around 50% reported to always complete homework on time, with a minority reporting less than always, with factors like time management (17% in Brussels, 25% in Seville) and material comprehension (56% in Brussels, 28% in Seville) affecting completion. Students' self-efficacy varied by task, with 30% of the students feeling they could "easily" write a technology essay (task 2) but only 6% and 11% feeling confident about explaining sound propagation (task 3) or solving a numerical estimation task such as guessing the litres of water in a pool (task 1). Thus, the sample was substantially homogeneous in terms of demographics but diverse in terms of self-efficacy.

Table 2 illustrates the distribution of student characteristics across treatment groups alongside the results of separate Fisher's Exact Tests on the association with treatment assignment. The sample characteristics were generally balanced across treatment groups. Only one variable out of ten – i.e., self-reported difficulties in homework – was statistically associated ($p = 0.05$) with the Socratic/Non-Socratic assignment, while no significant associations were found for the AI-reasoning assignment. Thus, only one out of twenty Fisher's exact tests showed a significant result, indicating a good balance across treatments.

4.2 Attitudes Towards AI

As illustrated in Figure 3, students expressed conflicting views on the role of AI in education. On the one hand, a majority felt that AI is often misused by students (65%) and potentially dangerous (57%). On the other hand, most students also anticipated that AI would enhance student learning in the future (59%) and contribute positively to society (65%). These findings suggest that most students in our sample are aware of the risks of misuse and safety with a prevailing optimism that AI could support educational growth and societal progress.

4.3 The Impact of AI Step-by-step Reasoning Exposure

Nearly all students revised their initial estimates (110 out of 122), with no significant difference between treatment groups (Fisher's test, $p = 0.9$). However, presenting students with AI-generated step-by-step reasoning may have influenced the accuracy of their revised estimates.

To address RQ1, we examined whether the treatment—showing step-by-step AI reasoning—affected how students used the AI's prediction, compared to seeing the prediction alone without any accompanying explanation. For each student $i = 1, \dots, 122$, we computed the absolute error as the absolute difference between student i 's guess, Guess_i , and the actual value of coins:

$$\text{Absolute Error}_i = | \text{Guess}_i - \text{Actual Value Coins} |,$$

where a smaller absolute error indicated greater accuracy.

Figure 4 illustrates that students' prediction accuracy was positively skewed in both treatment groups, with a greater accuracy for students exposed to AI reasoning. To estimate confidence intervals for the median difference in accuracy between the groups, we used a nonparametric bootstrap approach.⁹ This procedure involved resampling participants' absolute errors ($n = 1,999$) to build a distribution of medians for each group. The 5th and 95th percentiles of the resulting distribution were used to construct a 90% confidence interval for the difference in accuracy between groups. The interval ranged from 0 to 70, indicating a greater accuracy (lower error) for the students exposed to AI-generated reasoning (one-sided, $p < 0.05$).

To examine the impact of AI explanations on students' perceived credibility of AI predictions (RQ2), we analysed students' five-point ratings of the perceived accuracy of the AI estimated value of coins. We noticed that 53% of students who did not receive step-by-step reasoning considered the AI estimate as either "good" (39%) or "very good" (14%) compared to 43% of students exposed to the AI reasoning (19% and 24%, respectively). This gap of ten percentage points suggests that students viewed the AI as more accurate when the AI-generated guess was presented as a "black box." However, the association is not statistically significant (Fisher's test, $p = .27$), and even regression analysis, controlling for individual characteristics, showed no significant association (Figure 5).

Conversely, exposure to the AI step-by-step reasoning seemed associated with students' perception of the accuracy of the "human" guess — the average guess from 600 participants reported in the original study (Steiner 2015).¹⁰ While all groups received the same human and AI estimates, 64% of students in the control rated the human estimate as "poor" or "very poor" against 45% of students exposed to AI reasoning. Although this difference was insignificant (Fisher's exact test, $p = 0.15$), regression analysis, controlling for fixed differences across schools and across individuals, such as their gender, the initial guess of the value of coins, and experience with ChatGPT, showed a significant effect ($p < 0.05$) of the assignment to AI reasoning on students' perceptions of the human guess, as shown in Figure 5. This finding underscores the complex relationship between AI and learning. It suggests that students exposed to the AI's step-by-step reasoning may have identified minor errors, thus increasing their expectations of human accuracy, and those who viewed AI as a "black box" tended to underrate human estimates.

4.4 A Comparison of Socratic vs Non-Socratic AI

4.4.1 Student-AI Interactions

Figure 6 shows the students' message length and frequency in the Socratic AI and Non-Socratic AI groups, allowing us to compare the student-AI interactions across treatment groups. As shown in the figure, Socratic AI students exchanged a median of 20 messages, significantly more than the eight mes-

⁹We detected one outlier in the data, which was removed from the bootstrap analysis: one student provided an unusual and notably high guess of 6969. However, this value deviated substantially from both the overall distribution and the student's initial guess, suggesting a mistake rather than an estimate.

¹⁰Notably, this guess was \$596, exceeding the correct value (\$379.54) by about the same amount as the AI guess underestimated it (\$213).

sages by non-Socratic AI students (Wilcoxon test, $p < 0.01$). In addition, the Socratic AI's messages were significantly shorter, with a median of 42 words, compared to 123 for the non-Socratic tutor (Wilcoxon test, $p < 0.01$). Socratic students also used fewer words, with two peaks: one at ten and the other at one word. This evidence supports the hypothesis that the Socratic tutor encouraged more engagement and interaction, resembling a relatively more genuine student-tutor talk.

To complement these measures, we examined qualitative differences in the chats between students and the AI tutor, as illustrated by selected chat examples in the Appendix. This analysis highlights that the interactions were qualitatively different. In the Socratic condition, students had to actively think about the problem and provide input in order to progress. For example, when discussing the effects of social media, a student in the Socratic group was first prompted by the AI tutor to list positive effects, and then asked to identify negative ones. In contrast, a student in the non-Socratic group was simply presented with a list of both positive and negative effects by the AI tutor.

4.4.2 Students' Confidence in Their Responses

To test whether Socratic AI increased students' confidence in their answers (RQ4), we examined students' self-reported confidence levels at the end of each task. As shown in Figure 7, the observed differences in confidence between the treatment groups were minimal. Specifically, 25% of Socratic students reported feeling "very confident," and 33% reported feeling "confident" compared to 21% and 38%, respectively, among non-Socratic students. These differences were not statistically significant, even with regression analysis controlling for variation across students and tasks (Figure 8). Thus, our analysis found no evidence of treatment effects on students' confidence in their answers.

Further explorative regression analysis to examine potential treatment interactions shows that Socratic AI students with prior ChatGPT experience reported significantly less confidence ($p < 0.1$) than Non-Socratic AI students (see Figure 12). This explorative finding suggests that experienced users may perceive new or unconventional AI tutoring methods as less effective or even counterproductive, indicating that encouraging the use of such AI tutoring tools among experienced ChatGPT students can be challenging.

4.4.3 Perceived Helpfulness of AI Tutor

To test differences in students' perceptions of the AI tutor's helpfulness (RQ5), we examined students' ratings of helpfulness of the AI interaction at the end of each task. These ratings were on a five-point scale, from "Not at all helpful" to "Very helpful."¹¹

In both treatment groups, many students found interacting with the AI tutor "very helpful" or "helpful", with 56% of the non-Socratic and 44% of the Socratic students, as shown in Figure 9. However, the

¹¹Due to a coding issue, the scale used in Brussels and Seville differed slightly in the labels: Seville's students saw "Extremely helpful" whereas Brussels students saw "Very helpful." However, this issue has a minimal impact on the overall results, as results focus on the negative labels ("not at all helpful" or "not helpful") and they remain the same even after controlling for location effects in a regression. Additionally, open discussions held with students after the experimental session confirmed that they found the Socratic AI less helpful. This feedback, combined with the observed differences in negative labels, reinforces that the coding discrepancy has a minimal impact on the overall findings.

Socratic treatment group showed a bimodal distribution, with a substantial fraction of students (21%) finding the interaction “not at all helpful.” The association between perceived helpfulness and treatment assignment was statistically significant (Fisher’s exact test, $p = 0.025$), providing evidence that Socratic AI was less helpful to certain students. This association, however, was stronger for Task 1 and Task 3 than Task 2, suggesting an association between the perceived AI’s helpfulness and the type of task. Ordinal logistic regression accounting for individual student and task characteristics, including self-efficacy per task, revealed a statistically significant negative difference that corresponds to a drop of approximately 13 percentage points in perceived helpfulness associated with the Socratic AI, as illustrated in Figure 10. See Section A.2 for more details.

4.4.4 Learning Outcomes and Knowledge Retention

To evaluate the impact of learning (RQ3), we compared the correctness of responses to Task 3, which focused on sound propagation, and assessed knowledge retention using a follow-up question on the same topic without AI assistance. As shown in Figure 11, the use of AI significantly enhanced students’ response accuracy. Before interacting with the AI tutor, only 32% of students correctly answered that sound travels faster in water than air due to water’s higher density. After AI interaction, this percentage nearly doubled increasing to 68%. However, there was no significant difference in learning outcomes between the Socratic and non-Socratic AI approaches, as illustrated in the figure. Moreover, when presented with a follow-up question (asking in which medium sound travels fastest among gold, rubber, warm air, cold air, and water), only 18% of students responded correctly by selecting the denser material (i.e., gold), again with no significant difference across treatments. This result suggests two implications. Firstly, we found no evidence of Socratic AI improving learning. Secondly, our results seem to suggest a problem of limited retention of the insights obtained with AI assistance or difficulty in applying such learning to novel scenarios.¹²

¹²While it is true that gold is significantly denser than water or air, it is possible that some students did not fully understand this fact or were unsure how to compare the densities of the materials. If that were the case, even if the AI effectively conveyed that sound travels faster in denser media, students lacking this knowledge may still have been unable to select the correct answer. However, this explanation seems unlikely, as the relative densities of air, water, and metals, like gold, are commonly taught and conceptually straightforward, suggesting that other factors have contributed to students’ outcomes.

5 Discussion

This study contributes to our understanding of the impact of pedagogically-aligned configurations of an LLM-based tool on the learning and attitudes of high school students, with a special focus on their critical thinking.

5.1 Overview of the findings

With the rapid adoption of AI in education, the focus on students' competencies and skills, such as critical thinking, has become particularly urgent (Walter 2024; Holmes, Miao, et al. 2023). Despite the rapidly increasing body of research on the impact of LLMs on students' learning, most studies focus on participants in higher education (e.g., Suriano et al. 2025), which may not generate insights applicable to younger students with different cognitive maturity. In addition, despite the increasing research interest regarding the impact of LLMs on students' learning, there is little consensus among researchers, mainly due to students' over-reliance on AI tools (Zhai, Wibowo, and Li 2024).

The results of this study focus on the role of two interventions in configuring an LLM chatbot for educational support: (i) step-by-step explanations and (ii) the Socratic method with guiding questions. First, we found that students significantly benefit from AI-generated step-by-step reasoning accompanying solutions, mainly when performing open-ended tasks like estimating unknown quantities (RQ1). This finding aligns with expectations of current research on the use of LLM chatbots and their ability to provide explanations for learning purposes (Y. Wu 2024). It also aligns with previous research demonstrating that LLMs can enhance student performance (Xing et al. 2025). However, our results underscore that the key mechanism driving AI improvements is the step-by-step reasoning provided by the AI, which allows students to understand better and engage with problem-solving.

Furthermore, our study reveals that step-by-step reasoning not only helps students in solving problems but also enhances students' ability to evaluate AI-generated information critically (RQ2). Critical thinking is indeed a complex cognitive process that involves the evaluation and analysis of a given information (Lai 2011). The mobilisation of students' critical thinking in the specific task was evidenced by the more positive evaluations of human-generated guesses compared to AI-generated solutions, suggesting that students could better assess and challenge AI predictions. This contribution extends the existing literature by showing that AI can foster critical thinking and analytical skills when coupled with transparent reasoning rather than being presented as a "black box" (Bearman and Ajjawi 2023).

In addition, we compared various ways to structure student-AI interactions and, more specifically, the effectiveness of Socratic AI (an interactive, questioning-based AI) with non-Socratic AI. Our results showed that Socratic AI was more engaging and promoted greater interaction, aligning with the notion that AI-student interactions should be dynamic and dialogical (Suriano et al. 2025). However, contrary to our expectations, we found no significant differences in students' self-reported confidence in the accuracy of their answers or in the correctness of their responses despite the more frequent interactions with the Socratic AI (RQ4). Additionally, the Socratic AI's perceived helpfulness was rated lower compared to the

non-Socratic AI (RQ5). These results cast some doubts on the effectiveness of Socratic AI in short-term tasks or, more broadly, the effect of certain kinds of AI-student interactions. As such, existing pedagogical practices extensively used in human-human interaction might not always work in human-AI interaction. In addition, while recent research attempts to understand the quality of Socratic LLMs for critical thinking (e.g., [Favero et al. 2024](#)), these studies use simulations without experiments with human participants. Our results underscore the need for adapting existing pedagogical paradigms or proposing new ones for integrating LLM tools into pedagogical practices effectively.

Interestingly, our findings indicate that simply interacting with AI cannot promote meaningful and lasting learning (RQ3). In our initial test on sound propagation, approximately half of the students could revise their initial answers based on AI interactions, improving their response accuracy from 30% to 70%. However, in a verification task where students had no access to AI, the majority failed to identify the correct answer, mistakenly claiming that sound propagates faster in water than in gold. Most students exhibited this misunderstanding, which supports key concerns that AI-generated answers alone do not scaffold effective learning and the mere implementation of Socratic AI does not mitigate this risk.

Our results suggest that AI has great potential as an educational tool, but its implementation requires careful consideration. First, students with prior experience in using AI tools may not readily adopt new pedagogical approaches, especially if they perceive them as less effective than commercially available alternatives. Second, the effectiveness of the pedagogical approach may vary depending on the nature of the task, complicating the design of a one-size-fits-all solution for AI-assisted learning.

5.2 Limitations of the study and future work

Several limitations of our study should be acknowledged. The small sample size limits the generalizability of our findings, though the controlled environment and the use of multiple tasks help mitigate potential noise in the data. Also, our experiment focused on short-term results. Although short-term results are important to foster adoption, it is unclear the effectiveness of AI in the long term. Another limitation is that learning retention was tested using only one specific physics question. Although we controlled for prior knowledge and carefully designed a simple task to fit within a 40-minute intervention, additional questions would be needed to rule out confounding factors. Additionally, our study combined objective performance metrics with self-assessed ratings of helpfulness and confidence. However, individual perceptions do not necessarily reflect actual learning ([Noroozi et al. 2025](#)). Further research is needed to validate our findings using objective learning measures.

Finally, we conducted our study with a cohort of students possessing strong English proficiency and a clear understanding of the limitations of AI, which may not be representative of the general student population. Additionally, we focused on only two schools, which allowed us to control for school-specific fixed effects. However, we recognise that the impact of the treatments may differ across schools, and a broader investigation involving many more institutions would be necessary to explore this variability. Furthermore, the experiment was carried out at school and in a secure and anonymous digital environment, with a robust protocol developed to ensure the safe and ethical handling of AI-based interactions

in experimental settings. However, it remains to be seen if the results of our analysis will remain when students use AI in the field.

Our future work includes a large-scale study with a representative sample in a country in Europe, where we aim to address the above-mentioned limitations.

5.3 Concluding remarks

While our study focused on comparing a fairly general pedagogical approach—Socratic AI, future research should explore this pedagogical method in a more nuanced way and consider alternative pedagogical approaches to facilitate the scalability of AI tools across diverse tasks and educational contexts. Yet, our findings have important implications for designing AI tutors, highlighting the importance of providing transparent AI-generated step-by-step reasoning and the challenges of fostering learning through guided AI-student interactions. Therefore, AI systems must engage students interactively and provide learning opportunities for students critical thinking and problem-solving skills to maximise their educational value. Future developments should focus on refining the integration of AI-generated reasoning and ensuring that AI tools are adaptable to various learning tasks and students' needs.

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A**B**

Figure 1: Classrooms used for the experiment in Seville (**A**) and in Brussels (**B**).

A

50%


AI "Socratic"



User: I'm struggling to understand the concept of atoms. Can you help me?

Socratic AI: Of course! Let's start by exploring your current understanding. What do you already know about atoms?

50%


AI "non-Socratic"

B

50%


AI with explanation



User: What's the value of coins in the jar in the image?

AI: "The jar in the image resembles a standard large water bottle, possibly around 5 gallons (18.93 liters) in volume. Since the coins don't fill the jar completely, we estimate they occupy about 1/4 of the jar. [...]"

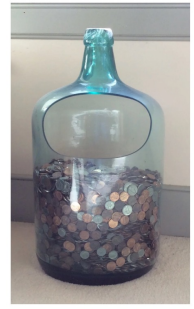
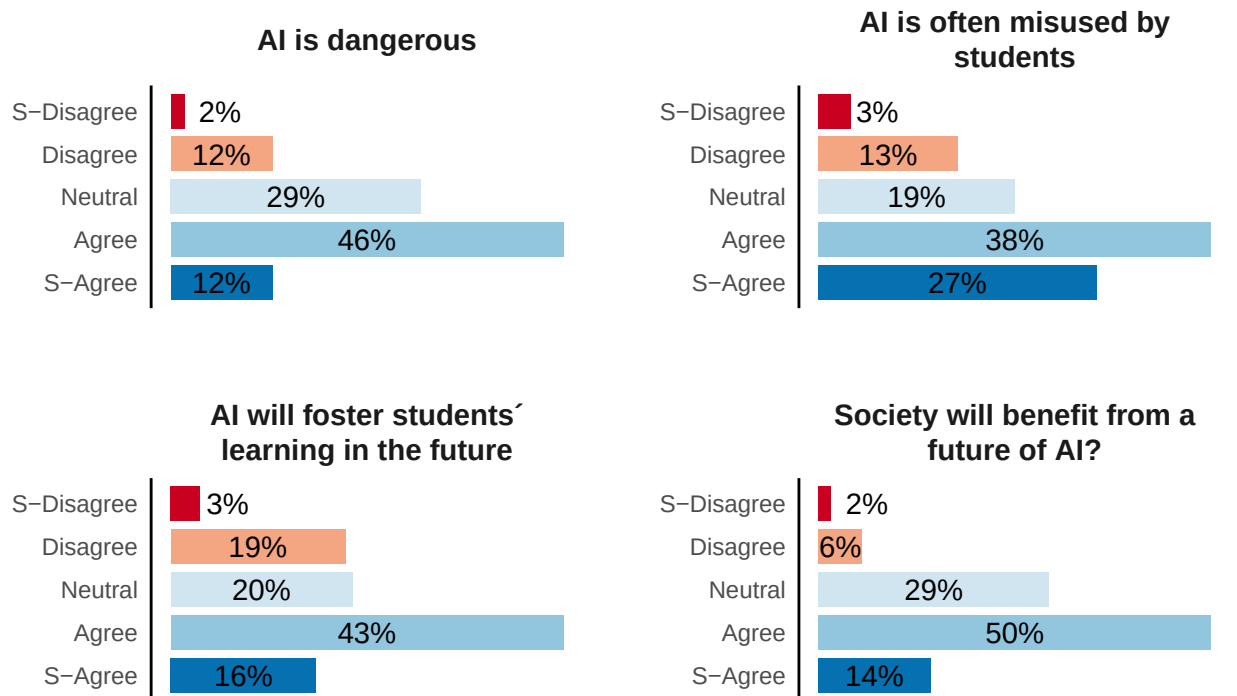


Figure 2: The experimental design involves two manipulations: **A.** Socratic vs Non-Socratic; **B.** AI solution with explanation vs AI solution (without explanation)

Student Attitudes on AI in Education

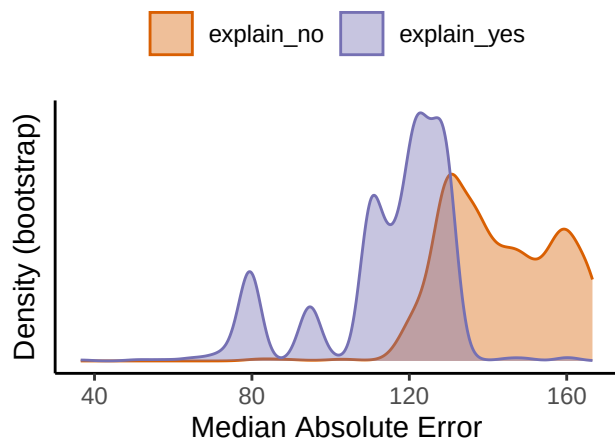
% of students who agree or disagree with the statement:



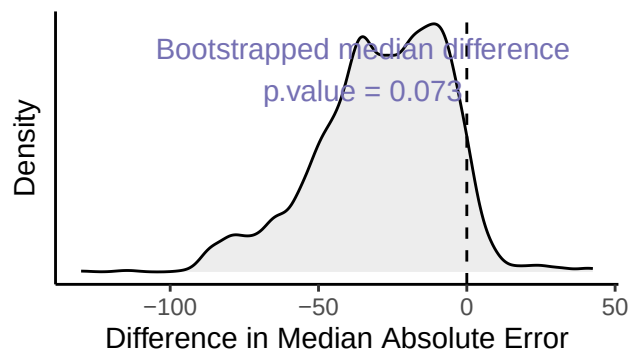
Source: Full sample (n = 122)

Figure 3: Student attitudes on AI in education

A



B



Source: Full sample (N = 121, excluding one outlier).

Figure 4: Comparison of Students' Absolute Error in Estimating Coin Jar Value with and without AI Explanations. All students received the AI-generated estimate of \$213 (the correct value was \$379.54). Still, those in the AI reasoning treatment also viewed the AI-generated step-by-step explanation for the estimation. Exposure to the AI-generated explanation significantly reduced the students' median absolute error.

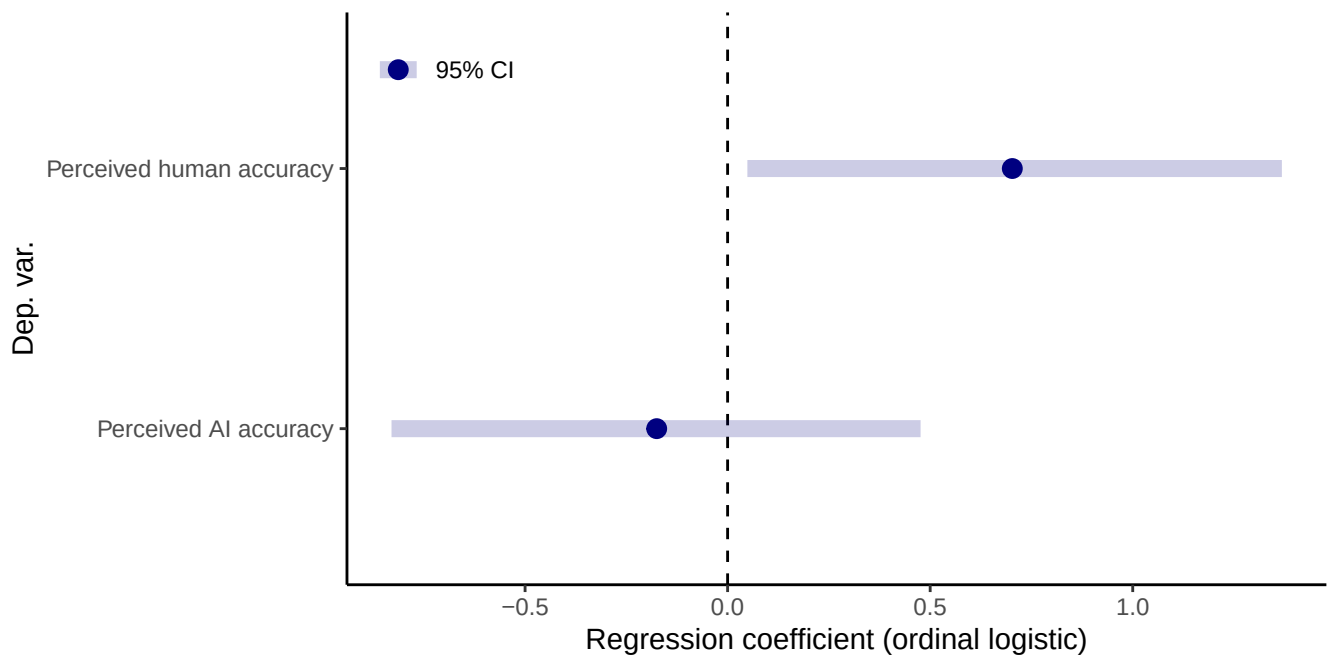


Figure 5: Comparison of students' perceived accuracy of AI and human estimates across treatments on a five-point scale. Coefficients from separate ordinal logistic regressions controlling for students' location, gender, and prior experience with ChatGPT. Positive coefficients indicate increased perceived accuracy associated with students' exposure to AI reasoning.

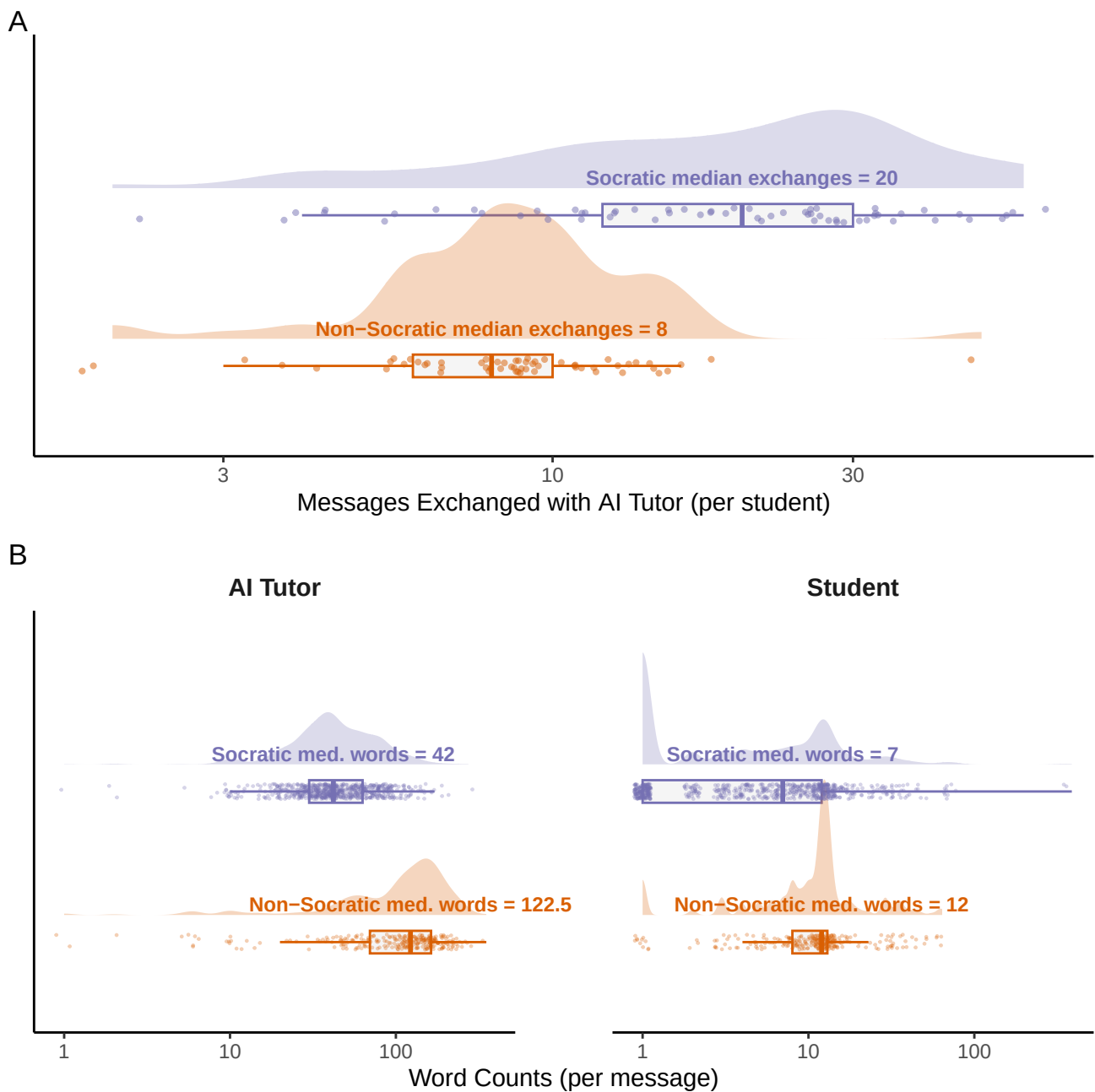
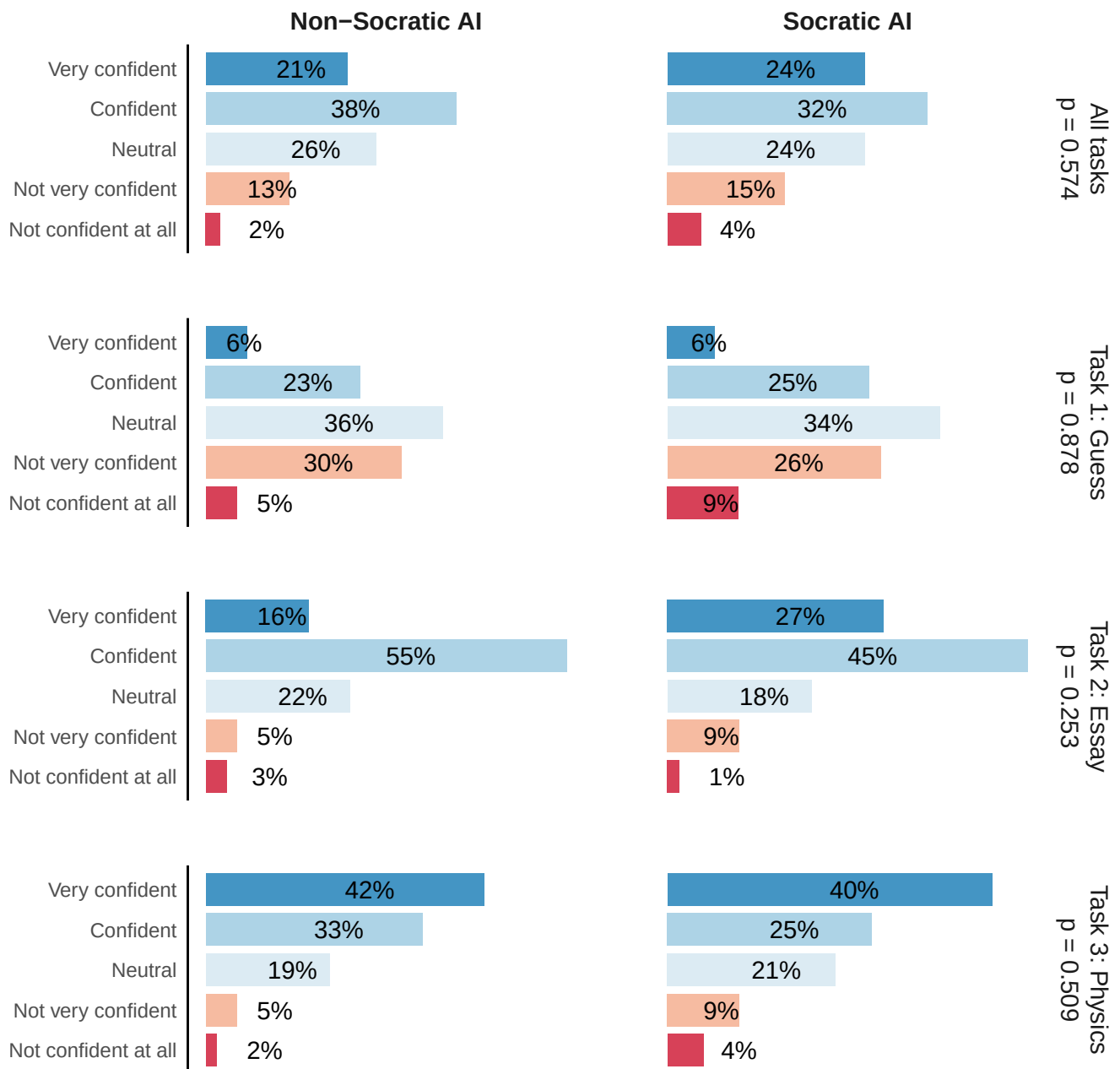


Figure 6: Comparison of message frequency and word count per message across Socratic and non-Socratic treatments. Top panel (**A**) shows differences in the frequency of messages exchanged, while the bottom panels (**B**) depict the word counts per message for the AI tutor (left) and the students (right).



Source: Full sample of student-task combinations (n = 364)

Figure 7: This figure shows the impact of Socratic AI on students' confidence levels in their performance across three different tasks. Despite some differences between the Socratic and Non-Socratic groups, we found no significant association between the treatment assignment and students' declared confidence levels.

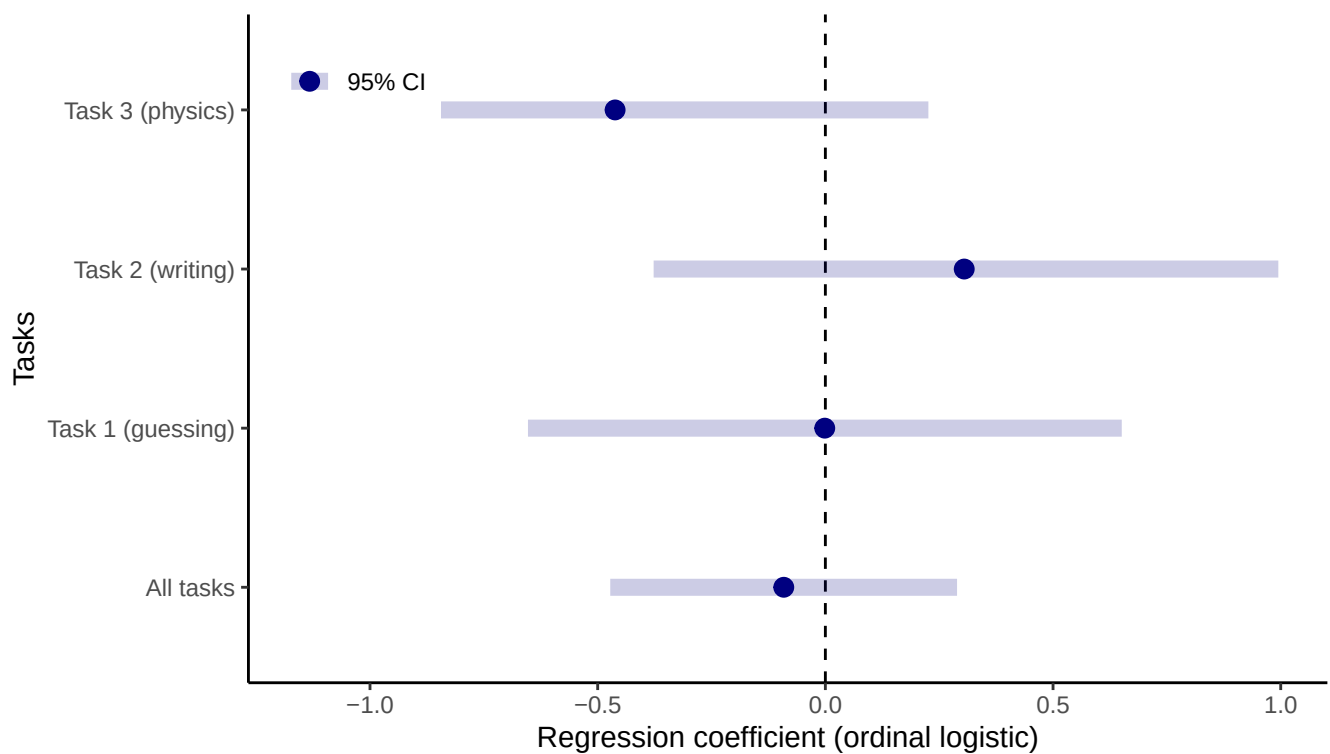
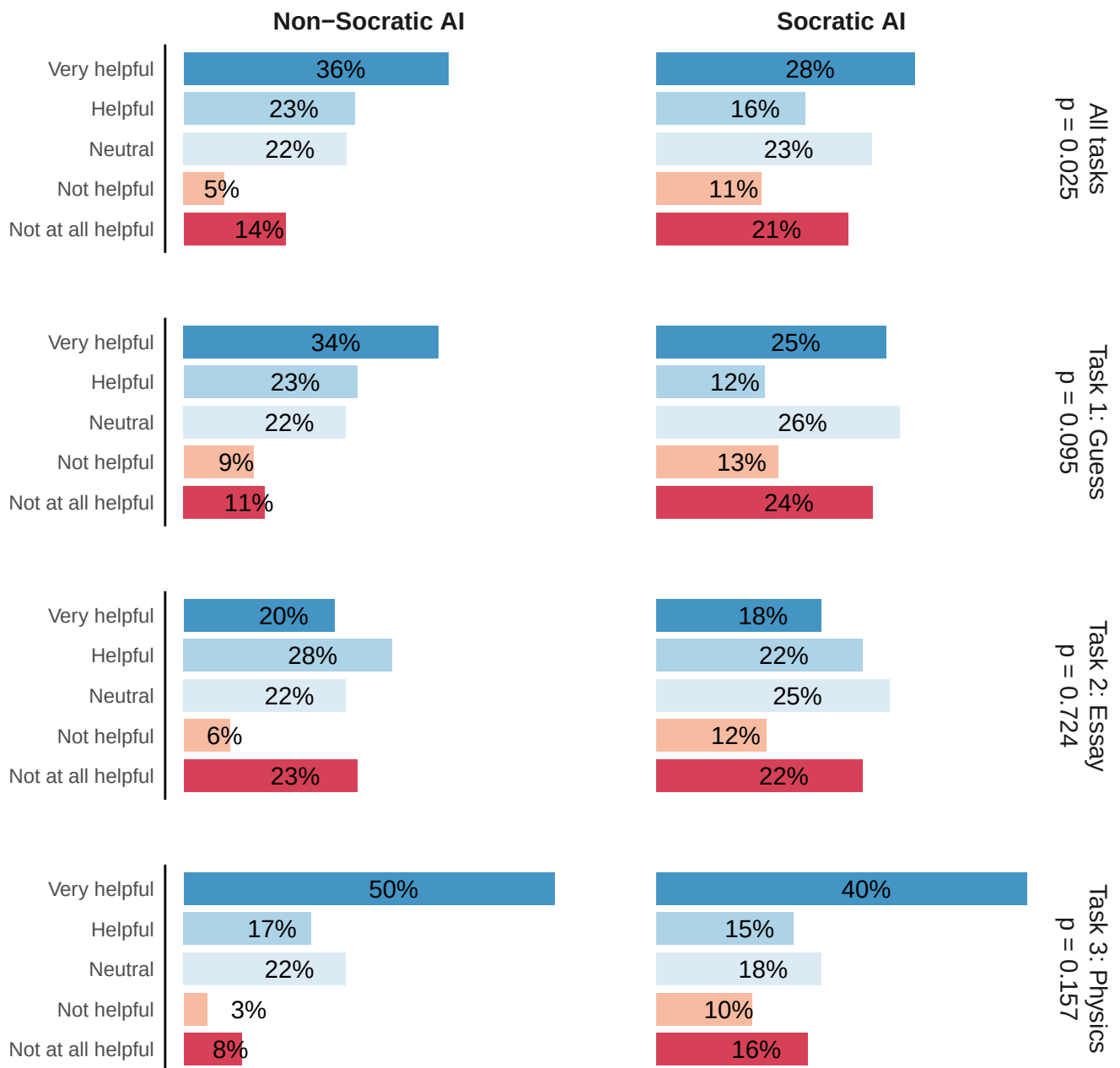


Figure 8: This figure shows the coefficients from separate ordinal logistic regressions on students' confidence in their answer on a five-point scale, controlling for students' self-efficacy per task and task fixed effect. Results show no significant association between the treatment assignment and students' confidence levels.



Source: Full sample of student-task combinations (n = 364)

Figure 9: This figure shows the impact of Socratic AI on students' perceived helpfulness of the AI tutor across three different tasks.

Effect of Socratic AI on Students’ Perceived Helpfulness of the AI Tutor

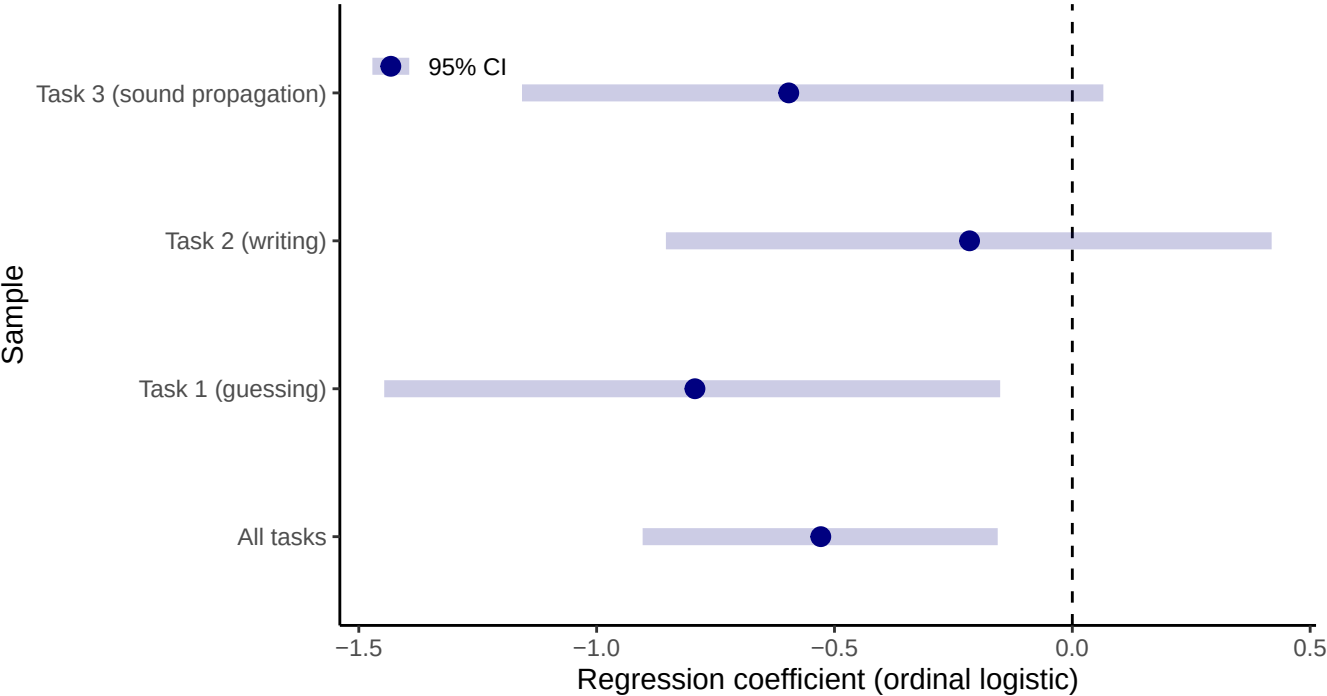
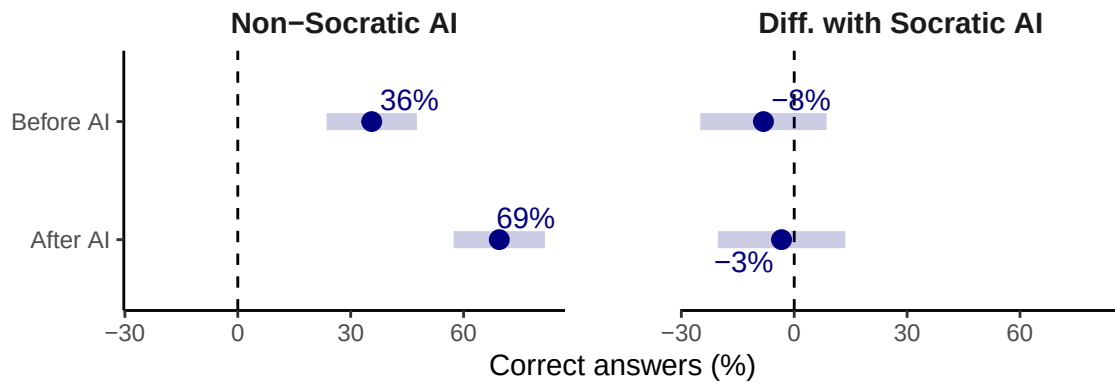


Figure 10: The impact of Socratic AI on students’ confidence levels in their performance across three different tasks. Despite some differences between the Socratic and Non-Socratic groups, we found no significant association between the treatment assignment and students’ declared confidence levels

A



B

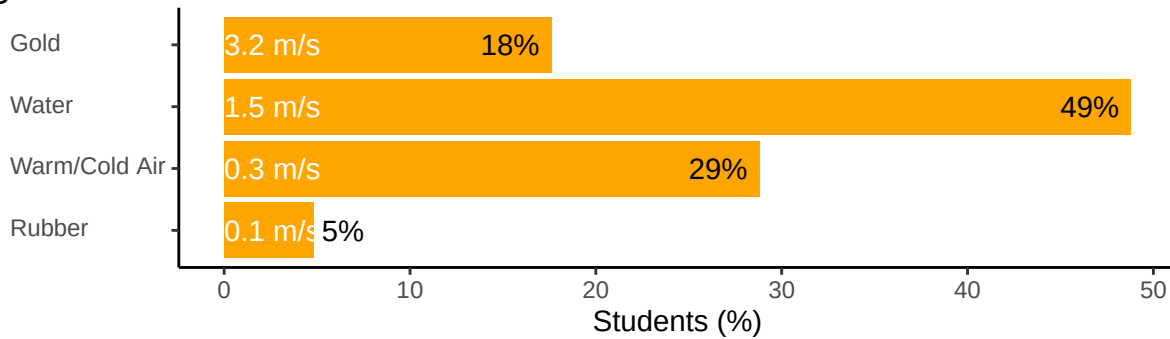
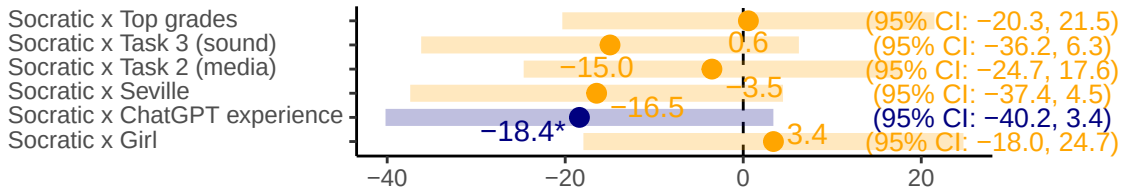


Figure 11: Effects on learning: **A** shows the % of correct responses about the physics of sound propagation, illustrating the positive effect of interacting with the AI tutor, while no differences are associated with the Socratic AI. **B** shows the results of the verification question asking students to identify the fastest material for sound propagation (speed as meter per second reported) without AI assistance. Only 18% responded accurately, indicating limited learning.

A

Socratic AI and Student Confidence Levels

Interaction effects between Socratic AI and task/student character



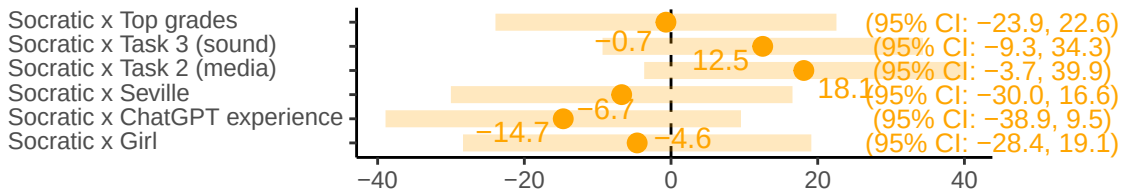
Regression coefficients and 95% confidence intervals

Results from linear regression (n = 366); * p < 0.1

B

Socratic AI and Perceived Helpfulness

Interaction effects between Socratic AI and task/student character



Regression coefficients and 95% confidence intervals

Results from linear regression (n = 366); * p < 0.1

Figure 12: Regression coefficients and 95% confidence intervals from a linear regression using as dependent variable the (A) students' self-reported confidence level and (B) perceived helpfulness of the AI tutor. The controls include student's gender, grades, location, and ChatGPT experience. Treatment dummies are interacted with all the controls and the task type. All models also include individual student random effects, and the student's rated confidence in their skills before performing each task.

Table 1: Prompt instructions associated with the AI Tutor's treatments

AI.Tutor	Prompt.instruction
Socratic	You are a Socratic tutor. You always answer using the Socratic style, asking just the right questions to help students learn to think for themselves and breaking down the problem into simpler parts until it's at the right level for them. You provide concise information and explanations understandable for 8th to 10th grade students.
Non-Socratic	You are a didactic tutor. You provide concise information and explanations understandable for 8th to 10th grade students.

Table 2: Sample Characteristics

Name	Value	Socratic (%)			AI Explain (%)			N
		Yes	No	p.	Yes	No	p.	
Gender	Boy	55	47	0.47	49	53	0.72	62
	Girl	45	53		51	47		59
Grades	Average or lower grades	47	45	0.86	43	48	0.58	55
	Top grades	53	55		57	52		65
Has used ChatGPT	No	43	37	0.58	34	45	0.27	49
	Yes	57	63		66	55		73
Homework difficulties	Clear instructions from teachers	10	20	0.05	14	16	0.53	18
	Effective time management	29	12		25	17		25
	Lack of distractions	18	10		9	19		17
	Support from family or peers	8	7		9	6		9
	Understanding the material	35	51		44	42		52
Homework on time	Always on time	42	56	0.33	56	42	0.27	59
	Sometimes or rarely on time	8	7		5	9		9
	Usually on time	50	37		39	48		53
Homework weekly study	0-1 hours	47	41	0.52	44	44	0.12	53
	1-2 hours	34	44		46	33		47
	2+ hours	19	15		11	23		21
Location	Brussels	49	56	0.47	50	55	0.72	64
	Seville	51	44		50	45		58
Self-efficacy (essay)	Could do easily	38	32	0.73	29	41	0.45	43
	Could do with a bit of effort	52	53		55	50		64
	I couldn't do this on my own	3	3		5	2		4
	I would struggle on my own	6	12		10	8		11
Self-efficacy (guess)	Could do easily	14	12	0.60	9	17	0.17	16
	Could do with a bit of effort	27	34		36	25		37
	I couldn't do this on my own	24	15		24	16		24
	I would struggle on my own	35	39		31	42		45
Self-efficacy (sound)	Could do easily	6	7	0.35	7	6	0.66	8
	Could do with a bit of effort	46	31		33	44		47
	I couldn't do this on my own	14	17		17	14		19
	I would struggle on my own	33	46		43	36		48

Table 3: % of students in each treatment group by responses to the question ‘Does sound travel faster in water than air?’

	Full sample	Non-Socratic	Socratic AI
Same speed	9.1	6.8	11.3
Water is faster because higher density (correct)	31.4	35.6	27.4
Water is faster because lower density	11.6	10.2	12.9
Water is slower because higher density	41.3	42.4	40.3
Water is slower because lower density	6.6	5.1	8.1

A Supporting Information

A.1 AI Explanation

Table [A1](#) presents the text shown to participants in the “AI with Explanation” treatment. This was generated by prompting GPT-4.0 with an image of a coin-filled jar with the following prompt: “What is your best guess of the total value of coins in the jar pictured in this image? Answer with a concise step-by-step procedure understandable for an 8th-grade student and a number representing the estimated total value.”

The image was from the article, “Turns Out the Internet Is Bad at Guessing How Many Coins Are in a Jar” ([Steiner 2015](#)). Notice that, since we couldn’t assume students were familiar with the U.S. coins, in the experiment we combined the jar picture with another image illustrating the coin denominations.

A.2 Regression analysis for differences in students confidence

We analyzed the impact of Socratic AI versus Non-Socratic AI on student i 's outcomes after task j , Y_{ij} , using different specifications based on following ordinal regression model:

$$P(Y_{ij} = k) = \beta_0 + \beta_1 \text{self-efficacy}_{ij} + \gamma T_i + \delta_j + \eta_i + \epsilon_{ij}.$$

Where:

- $P(Y_{ij} \leq k)$ represents the cumulative probability for Y_{ij} and $P(Y_{ij} = k)$ is the probability that the self-reported confidence level or perceived helpfulness of student i for task j falls at a certain point k .
- $\text{self-efficacy}_{ij}$ is the student i 's self-efficacy relevant for task j (i.e., perceived ability to perform the task)
- T_i indicates the AI tutor assigned to student i (1 = Socratic or 0 = non-Socratic).
- δ_j is a random effect associated with task j , accounting for task-specific variation.
- η_i is a random effect associated with student i , capturing individual-level differences.
- ϵ_{ij} is the error term.

The regression specification described above assumes homogeneous treatment effects across individuals and tasks. To relax this assumption, we employed a more flexible model that allows the average treatment effect, γ , to vary based on individual-level and task-specific factors. Specifically, we included interaction terms between the treatment indicator and variables such as the student's gender, academic performance (as measured by school grades), prior experience with ChatGPT, and school location. Additionally, we accounted for heterogeneity in treatment effects across tasks by estimating separate regressions for each task-specific subset of the data. The results are shown in Figure 12, Figure 8, and Figure 10.

“Explanations are a central component of learning, both in everyday cognition and formal education. Research suggests that generating explanations, especially self-explaining, can be more effective for learning and generalization than simply reading text or receiving feedback (Lombrozo 2006). In educational settings, explanations are widely used across contexts, such as classroom instruction (Ma 1999), tutoring (Chi et al. 2001), peer learning (Webb 1989), and learning from texts (Goldman 2004), but their effectiveness depends heavily on how they align with learners' prior knowledge, focus on conceptual understanding over rote procedures, support active learning, and do not replace the learner's own knowledge construction (Wittwer and Renkl 2008).”

“AI-generated explanations, however, often fall short on these dimensions: they may not adapt to individual knowledge levels, ignore learner agency, and risk fostering overreliance due to the opaque, “black-box” nature of models, potentially spreading misinformation or suppressing

critical thinking (Rudin et al. 2022). Nonetheless, chatbots offer a unique opportunity for adaptivity, allowing learners to engage in Socratic dialogue and refine their understanding through follow-up questions (Yan, Greiff, et al. 2024). Our study explores this duality by first examining the impact of non-adaptive AI explanations and then testing whether adaptive, Socratic-style interactions better support learning compared to traditional classroom explanations.”

“Building on this, adaptive learning via AI chatbots can be conceptualized as a form of *collaborative learning* between the student and the AI tutor (Dillenbourg 1999). Collaboration, as defined by Dillenbourg, encompasses four aspects: the learning situation, the nature of interactions, the underlying cognitive mechanisms, and the outcomes for learners. In this study, we focus on the second aspect—*the interactions*—to investigate how different dialogic strategies employed by a LLM shape students’ learning experiences and perceptions. By framing the AI-student exchange as a collaborative process, we aim to understand how the structure and responsiveness of chatbot explanations, especially those modeled on Socratic questioning, can promote deeper engagement, conceptual understanding, and critical reflection compared to more static or directive approaches.”

B Acknowledgements

The contents of this publication do not necessarily reflect the position or opinion of the European Commission or Joint Research Centre. We thank Lucas Bagnari de Seabra and the colleagues in the JRC, especially those in the Digital Economy unit for their support and generous collaboration. Special thanks for the Data Privacy Coordinator of the JRC, who helped us with the data privacy statement, ensuring the highest safeguard to the students participating in the study. Many thanks go to the schools, including their leadership teams, parents and pupils who participated in the study. All errors are our own.

C Statements and Declarations

C.1 Data Availability

The data that support the findings of this study are available in the following open data repository [DOI available upon publication]. All relevant data have been anonymized to comply with ethical guidelines and institutional policies.

C.2 Ethical approval

This study was conducted in accordance with the ethical standards and guidelines of the European Commission. Ethical approval was obtained from the European Commission's Ethical Review Board prior to the commencement of the research. Approval reference number: 32530 / Ares(2023)3278765. All procedures involving human participants/data complied with the relevant institutional and European ethical regulations. Clearance received on May 17, 2023.

C.3 Informed consent for Participation

Informed consent was obtained prior to the experimental sessions. The text of the informed consent is provided below.

Purpose of the study

The purpose of this study is to understand the ways teachers and students use applications based on Generative AI, such as the ChatGPT, and the impact of this emerging technology on teachers and students' learning process. The findings of this study will help us understand how to increase the positive impact of the use of Generative AI while mitigating possible risks. This study aims to contribute to the current scientific dialogue about the use of Generative AI in Education and the teaching practices and provide recommendations for policymakers and developers in line with the European charter of fundamental rights and children's rights in the context of digital and online world.

Consent

I, the undersigned (Participant's name), born on (Birthdate) in (Birthplace), having read the related privacy statement and information brochure of the study "The impact of Generative Artificial Intelligence (AI) on Education: The case study of ChatGPT in Secondary Education", organised by the European Commission, Joint Research Centre, which is attached to the invitation hereby:

(1) Processing of my personal data for my participation in the study:

- ☐ Consent to the processing of my personal data, as described in the attached privacy statement, for the purpose of participating in this study.
- ☐ DO NOT consent to the processing of my personal data for the purpose of participating in this study.

(2) Processing of my personal data for research purposes:

- ☐ Consent to the processing of my personal data, including my recordings and related artifacts, as described in the attached privacy statement, for the purpose of allowing the research team to conduct research associated to this study and its objectives.
- ☐ DO NOT consent to the processing of my personal data, including my recordings and related artifacts, for the purpose of allowing the research team to conduct research associated to this study and its objectives.

(3) Anonymisation of my personal data for research and scientific purposes (only if you selected YES to consent request number 2 above)

- ☐ Consent to the anonymization of my personal data, including audio recordings and related artifacts from the study, to be used for educational and scientific purposes such as conference publications and presentations in an anonymous way.
- ☐ DO NOT consent to the anonymization of my personal data, including audio recordings and related artifacts from the study.

Contact Information

If you would like to exercise your rights under Regulation (EU) 2018/1725, or if you have comments, questions or concerns, or if you would like to submit a complaint regarding the collection and use of your personal data, please feel free to contact the Data Controller at:
EMAIL

C.4 Competing Interests

The authors declare no competing interests directly or indirectly related to the work submitted for publication.

C.5 Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used Grammarly, ChatGPT in order to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

D Examples of Student-AI Interactions

D.1 Estimation task

Question: How much water in litres do students consume at our school each week? Interact with the AI tutor at the bottom of this page before answering.

D.1.1 Non-Socratic Approach

Student: how many litres of water do we consume at school each week?

AI Tutor: To answer this question, we need more specific information. The amount of water consumed at a school each week can vary greatly depending on several factors: 1. Number of students and staff: More people means more water usage for drinking, washing hands, and other activities. 2. School facilities: If the school has a cafeteria, gym, or other facilities that use water, this will increase the total consumption. 3. Drinking habits: If students and staff are encouraged to drink a lot of water, this could also increase the total consumption. For example, if a school has 500 students and each student drinks an average of 1 litre of water per day, the school would consume 2,500 litres of water in a 5-day week just for drinking. This doesn't include water used for other purposes like cleaning or cooking. Remember, this is a very rough estimate and the actual amount could be much higher or lower depending on the factors mentioned above.

Student: if there is 5000 children at our school how many litres do we use per week?

AI Tutor: If we use the same estimation as before, where each student drinks an average of 1 litre of water per day, then a school with 5,000 students would consume 25,000 litres of water in a 5-day week just for drinking. However, this is a rough estimate and does not include water used for other purposes like cleaning, cooking, or facilities like bathrooms and gyms. The actual amount could be much higher. It's also important to note that promoting good hydration is essential for health and learning, so schools should encourage students to drink enough water each day.

D.1.2 Socratic Approach

Student: How much water in litres do students consume at our school each week?

AI Tutor: To answer that question, we need to gather some information first. 1. How many students are there in your school? 2. On average, how many litres of water does each student consume per day? 3. How many days per week are students at school?

Student: 1. There are around 200 people at our school

AI Tutor: Great, that's a good start. Now, let's move on to the next questions: 2. On average, how many litres of water does each student consume per day? 3. How many days per week are students at school?

Student: 2. 250 litres

AI Tutor: That seems like a lot of water for one person to consume in a day. The average person usually drinks about 2 litres of water per day. Could you please check that number again?

Student: 2 litres

AI Tutor: Great! Now, let's move on to the last piece of information we need. 3. How many days per week are students at school?

Student: 4 days, but on Wednesdays we don't consume water at school

AI Tutor: Okay, so if students are at school for 5 days a week, but they don't consume water at school on Wednesdays, that means they are consuming water at school for 4 days a week. Is that correct?

Student: Yes

AI Tutor: Great! Now, let's put all this information together. If there are 200 students in your school, and each student consumes 2 litres of water per day, and they do this for 4 days a week, how many litres of water do students consume at your school each week?

Student: 1600 litres in total

AI Tutor: That's correct! If each of the 200 students consumes 2 litres of water per day for 4 days a week, that would indeed total to 1600 litres of water consumed at your school each week. Well done!

D.2 Scientific question

Question: Does sound travel faster in water than air? And if so, why?

1. Sound travels slower in water due to its higher density
2. Sound travels faster in water due to its higher density [correct answer]
3. Sound travels at the same speed in both water and air, regardless of density
4. Sound travels slower in water due to its lower density
5. Sound travels faster in water due to its lower density

Now, interact with the AI tutor at the bottom of this page before answering. [SOCRATIC / NON-SOCRATIC] [Repeat question 21.]

D.2.1 Non-Socratic approach

Student: does sound travel faster in water or air and why?

AI Tutor: Sound actually travels faster in water than it does in air. This is because water has a higher density and elasticity than air, which allows sound waves to travel more quickly. In air, sound travels at about 343 meters per second, while in water, it travels at approximately 1,500 meters per second.

D.2.2 Socratic Approach

Student: Does sound travels faster in water than air? And if so, why?

AI Tutor: Yes, sound does travel faster in water than in air. But why do you think that might be the case? What do you know about the properties of air and water that might affect the speed of sound?

Student: does Sound travels faster in water due to its higher density?

AI Tutor: That's a great start! Density is indeed a factor. Sound waves travel through the vibration of atoms and molecules in a medium. In water, the molecules are much closer together than they are in air, which allows the vibrations to be transferred between them more quickly. But there's another property of water that also affects the speed of sound. Can you think of what that might be?

D.3 The effect of Social Media on Students

Question. Experts say that social media can have either a positive or a negative impact on students' well-being. What is your opinion about the effect of social media on teenagers? [...] Now, interact with the AI tutor at the bottom of this page before answering.

D.3.1 Non-Socratic

Student: what are the benefits of social media to teenagers

AI Tutor: 1. Communication and Connectivity: Social media platforms allow teenagers to stay connected with friends and family, even those who live far away. They can share experiences, updates, and communicate instantly. 2. Information and Learning: Social media can be a great source of information and learning. Many educational pages and groups provide knowledge about various subjects. It can also help them to stay updated with current events and news. 3. Self-Expression: Teenagers can use social media to express their thoughts, ideas, and creativity. They can share their artwork, music, or writings with a larger audience. 4. Community Building: Social media allows teenagers to join groups of people with similar interests. This can help them to feel a sense of belonging and community. 5. Skills Development: Managing social media profiles can help teenagers develop digital literacy and marketing skills. They learn about creating engaging content, online etiquette, and internet safety. 6. Emotional Support: Social media can provide a platform for teenagers to share their feelings and experiences, and receive emotional support. They can connect with others who are going through similar experiences. 7. Advocacy: Teenagers can use social media to advocate for causes they believe in and raise awareness about important issues. Remember, while social media has many benefits, it's also important to use it responsibly and be aware of potential risks such as cyberbullying, privacy issues, and misinformation.

Student: does it have more of a positive or negative impact

AI Tutor: The impact of social media on teenagers can be both positive and negative, and it largely depends on how it's used. On the positive side, social media can provide a platform for self-expression, learning, and connectivity. It can help teenagers develop digital literacy skills, find communities of like-minded individuals, and stay informed about current events. On the negative side, excessive use of social media can lead to issues like cyberbullying, privacy concerns, and exposure to inappropriate content. It can also contribute to feelings of inadequacy or anxiety, especially when teenagers compare their lives

to the often idealized representations they see online. In conclusion, social media can have both positive and negative impacts on teenagers. It's important for them to be educated about responsible use of these platforms, and for parents or guardians to monitor their online activities to ensure a healthy balance.

Student: thanks

D.3.2 Socratic Approach

Student: is social media a positive or a negative impact on student well-being?

AI Tutor: That's a great question and it's a topic that can be viewed from many different angles. Let's break it down. Firstly, can you think of any positive impacts that social media might have on a student's well-being?

Student: they can learn new things from social media

AI Tutor: Absolutely, that's a great point. Social media can be a source of information and learning. It can help students to connect with educational resources, learn about different cultures, or stay updated on current events. Now, can you think of any potential negative impacts that social media might have on a student's well-being?

Student: they use it too much

AI Tutor: Yes, that's a valid point. Excessive use of social media can lead to issues like distraction from studies, lack of physical activity, and even sleep deprivation. It can also contribute to feelings of anxiety or depression in some cases. Now, considering these positive and negative aspects, how do you think a balance can be achieved in using social media for the benefit of students?

E Questionnaire

1. Do you agree or disagree that society will benefit from a future of AI?

- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

2. Do you agree or disagree that AI is dangerous?

- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

3. Do you agree or disagree that AI will foster students' learning in the future?

- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

4. Do you agree or disagree that AI is often misused by students?

- Strongly Agree
- Agree
- Neutral
- Disagree
- Strongly Disagree

5. How easy or difficult would it be for you to explain how sound waves transfer in the air or other materials?

- I could do this easily
- I could do this with a bit of effort
- I would struggle to do this on my own
- I couldn't do this on my own

6. How easy or difficult would it be for you to write about the impact of technology on teenagers' well-being?

- I could do this easily

- I could do this with a bit of effort
 - I would struggle to do this on my own
 - I couldn't do this on my own
7. How easy or difficult would it be for you to guess how many litres of water are in an Olympic swimming pool?
- I could do this easily
 - I could do this with a bit of effort
 - I would struggle to do this on my own
 - I couldn't do this on my own
8. You see a jar filled with different coins; you must guess the total value of coins in US dollars.
9. We asked the same question about the jar's coins value to an AI system that can understand and process visual information. The AI guessed \$213. How accurate is this guess? [IF in AI EXPLANATION TREATMENT, ADD EXPLANATION HERE]
- Correct
 - Mostly Correct
 - Partially Correct
 - Mostly incorrect
 - Incorrect
10. We asked the same question to a group of 600 people of various backgrounds. People's mean guess was \$596. How accurate is this guess?
- Correct
 - Mostly Correct
 - Partially Correct
 - Mostly incorrect
 - Incorrect
11. Given the information from the AI (\$213) and the people's average (\$596), what is your final guess of the value of coins in the jar?
12. [Task 1] How much water in litres do students consume at our school each week? Interact with the AI tutor at the bottom of this page before answering. [Randomly assign SOCRATIC / NON-SOCRATIC AI Tutor]
13. How confident are you that the answer you provided is accurate?
- Very confident
 - Confident
 - Neutral
 - Not very confident
 - Not confident at all

14. How helpful was interacting with the AI tutor?
- Very helpful
 - Helpful
 - Neutral
 - Not very helpful
 - Not helpful at all
15. [Task 2.] Experts say that social media can have either a positive or a negative impact on students' well-being. What is your opinion about the effect of social media on teenagers? There is no correct or wrong answer.
- Very Positive
 - Positive
 - Neutral
 - Negative
 - Very Negative
16. Now, interact with the AI tutor at the bottom of this page before answering. [SOCRATIC / NON-SOCRATIC]. [Repeat question 15.]
- Very Positive
 - Positive
 - Neutral
 - Negative
 - Very Negative
17. Write a well-reasoned 600-character essay critically examining this topic. Write an introductory paragraph with background information, argumentation with as many convincing arguments or facts as possible, and a brief conclusion.
18. How confident are you that the arguments or facts in your essay are accurate?
- Very confident
 - Confident
 - Neutral
 - Not very confident
 - Not confident at all
19. How helpful was interacting with the AI tutor before writing the essay?
- Very helpful
 - Helpful
 - Neutral
 - Not very helpful
 - Not helpful at all

20. Can you explain why this interaction was helpful or wasn't? [TEXT]
21. [Task 3] Does sound travel faster in water than air? And if so, why?
1. Sound travels slower in water due to its higher density
 2. Sound travels faster in water due to its higher density [correct answer]
 3. Sound travels at the same speed in both water and air, regardless of density
 4. Sound travels slower in water due to its lower density
 5. Sound travels faster in water due to its lower density
22. Now, interact with the AI tutor at the bottom of this page before answering. [SOCRATIC / NON-SOCRATIC] [Repeat question 21.]
23. How confident are you that the answer you provided is accurate?
- Very confident
 - Confident
 - Neutral
 - Not very confident
 - Not confident at all
24. How helpful was interacting with the AI tutor?
- Very helpful
 - Helpful
 - Neutral
 - Not very helpful
 - Not helpful at all
25. [Verification question] In which of the following materials does sound travel faster?
- Rubber
 - Cold Air
 - Warm Air
 - Gold
 - Water [correct answer]
26. What is your average grade at school?
27. How many hours per day do you spend completing homework assignments?
28. How often do you complete your homework assignment on time?
29. What single factor contributes the most to your ability to complete homework assignments on time?
30. Have you ever used ChatGPT before today?
31. In your opinion, how many of your classmates are using ChatGPT for homework?

Table A1: AI's Step-by-Step Reasoning for Estimating the Value of Coins in a Jar

Step	Explanation
Introduction	To estimate the total value of the coins in the jar, let's go through a step-by-step process.
Estimate the volume of the jar	The jar in the image resembles a standard large water bottle, possibly around 5 gallons (18.93 liters) in volume. Since the coins don't fill the jar completely, we estimate they occupy about 1/4 of the jar. Therefore, the volume filled by coins is approximately 1.25 gallons or about 4.73 liters.
Estimate the volume of a single coin	Assuming most coins are US pennies, with a volume of around 0.36 cubic centimeters, we need to determine how many pennies could fit into the 4.73-liter volume (4730 cubic centimeters).
Convert to dollars	Finally, we convert the total number of pennies to dollars: $21350 \text{ pennies} \div 100 = \213.50 . Therefore, a rough estimation of the total value of the coins in the jar might be around \$213.50.